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Experiment 14: Decision Tree Classification

Aim

To implement the Decision Tree Classification algorithm using Python and classify data based on the given features.

Procedure

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree Classifier.
5. Train the model using the training data.
6. Predict the output for the test data.
7. Calculate the accuracy of the model.
8. Display the results.

Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target
```

```
# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Create model

model = DecisionTreeClassifier()
```

```
# Train model

model.fit(X_train, y_train)
```

```
# Prediction

y_pred = model.predict(X_test)
```

```
# Accuracy

accuracy = accuracy_score(y_test, y_pred)
```

```
print("Predicted Values:", y_pred)
```

```
print("Accuracy:", accuracy)
```

The screenshot shows a code editor with Python 3 selected. The code implements a Decision Tree Classifier using sklearn. The code is as follows:

```
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.tree import DecisionTreeClassifier
4 from sklearn.metrics import accuracy_score
5
6 # Load dataset
7 iris = load_iris()
8 X = iris.data
9 y = iris.target
10
11 # Split dataset
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.3, random_state=42
14 )
15
16 # Create model
17 model = DecisionTreeClassifier()
18
19 # Train model
20 model.fit(X_train, y_train)
21
22 # Prediction
23 y_pred = model.predict(X_test)
24
25 # Accuracy
26 accuracy = accuracy_score(y_test, y_pred)
27
28 print("Predicted Values:", y_pred)
29 print("Accuracy:", accuracy)
```

The right sidebar shows the execution results:

- Run** button and **Visualize Code** link.
- Enter Input here** section.
- Output** section.
- Status**: Runtime error.
- Time**: 0.0400 secs, **Memory**: 11.476 Mb.
- Runtime Error** section showing **SIGHUP**.
- Error** section.

Result

The Decision Tree Classification model was implemented successfully, and the predicted values were obtained with high classification accuracy.

Outcome

The experiment demonstrated how a Decision Tree can classify data efficiently using supervised machine learning.

Experiment 15: Feed Forward Neural Network (FFNN)

Aim

To implement a Feed Forward Neural Network using Python for classification.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing data.
4. Create a Feed Forward Neural Network model.
5. Train the model.
6. Predict the test data.
7. Evaluate the model accuracy.
8. Display the results.

Python Code

```
from sklearn.datasets import load_iris  
from sklearn.model_selection import train_test_split
```

```
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
# Load dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Create FFNN model
```

```
model = MLPClassifier(hidden_layer_sizes=(10,),
                      max_iter=1000,
                      random_state=42)
```

```
# Train model
```

```
model.fit(X_train, y_train)
```

```
# Prediction
```

```
y_pred = model.predict(X_test)
```

```
# Accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Predicted Values:", y_pred)
```

```
print("Accuracy:", accuracy)
```

```
Python3
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.neural_network import MLPClassifier
4 from sklearn.metrics import accuracy_score
5
6 # Load dataset
7 iris = load_iris()
8 X = iris.data
9 y = iris.target
10
11 # Split dataset
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.3, random_state=42
14 )
15
16 # Create FFNN model
17 model = MLPClassifier(hidden_layer_sizes=(10,),
18                       max_iter=1000,
19                       random_state=42)
20
21 # Train model
22 model.fit(X_train, y_train)
23
24 # Prediction
25 y_pred = model.predict(X_test)
26
27 # Accuracy
28 accuracy = accuracy_score(y_test, y_pred)
29
30 print("Predicted Values:", y_pred)
31 print("Accuracy:", accuracy)
```

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Runtime error

Time:	Memory:
0.0300 secs	11.736 Mb

Runtime Error

SIGHUP

Error

Traceback (most recent call last):

Result

The Feed Forward Neural Network was successfully trained and tested, producing accurate predictions on the test dataset.

Outcome

The experiment demonstrated the implementation of a Feed Forward Neural Network for solving a classification problem using Python.

These sections are ready to copy into a **Microsoft Word (.docx)** lab record.